

Lab Number: 03



Date: 2025/07/01

Title: OS Installation & Practice on Basic Networking Commands

THEORY:

Linux:

Linux is a family of open-source operating systems, based on the Linux kernel, first released in 1991 by Linus Torvalds. It's known for its flexibility, security, and wide range of applications, from smartphones and servers to supercomputers and embedded systems. While often referred to as an OS, Linux is technically a kernel, which is the core of an operating system. Linux distributions, like Ubuntu or Fedora, bundle the kernel with other software to create a complete OS.



Fig: Linux OS and its Evolution and Future

Virtual Box:

VirtualBox is free and open-source virtualization software that allows users to run multiple operating systems on a single physical machine. It enables the creation and management of virtual machines (VMs), each capable of hosting a different OS alongside the host system. This capability is useful for testing software, developing applications on various platforms, and creating isolated environments for experimentation. Essentially, it provides the functionality of multiple computers within a single computer.



Fig: Virtual Box

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VMware:

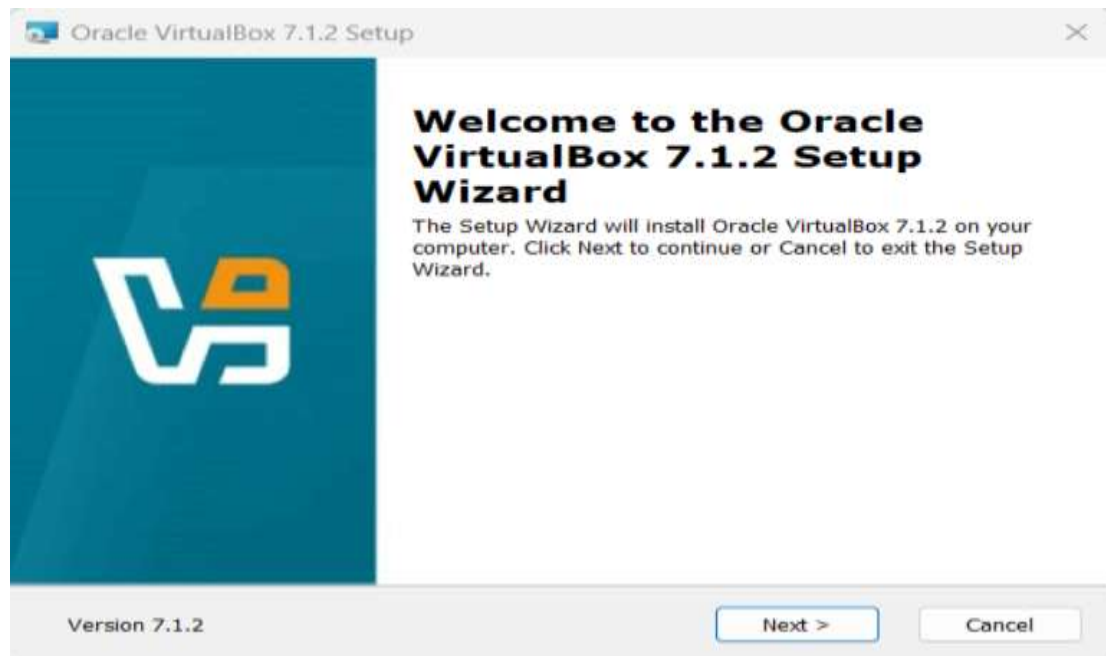
VMware is a leading provider of virtualization and cloud computing software and services. It enables the creation of virtual machines, which are software-based versions of physical computers, allowing users to run multiple operating systems and applications on a single physical machine. This virtualization technology offers benefits like increased efficiency, improved resource utilization, and simplified data center management. VMware's core product, vSphere, is a suite of virtualization tools that includes the ESXi hypervisor, which manages the allocation of hardware resources to virtual machines.



Fig: VMware

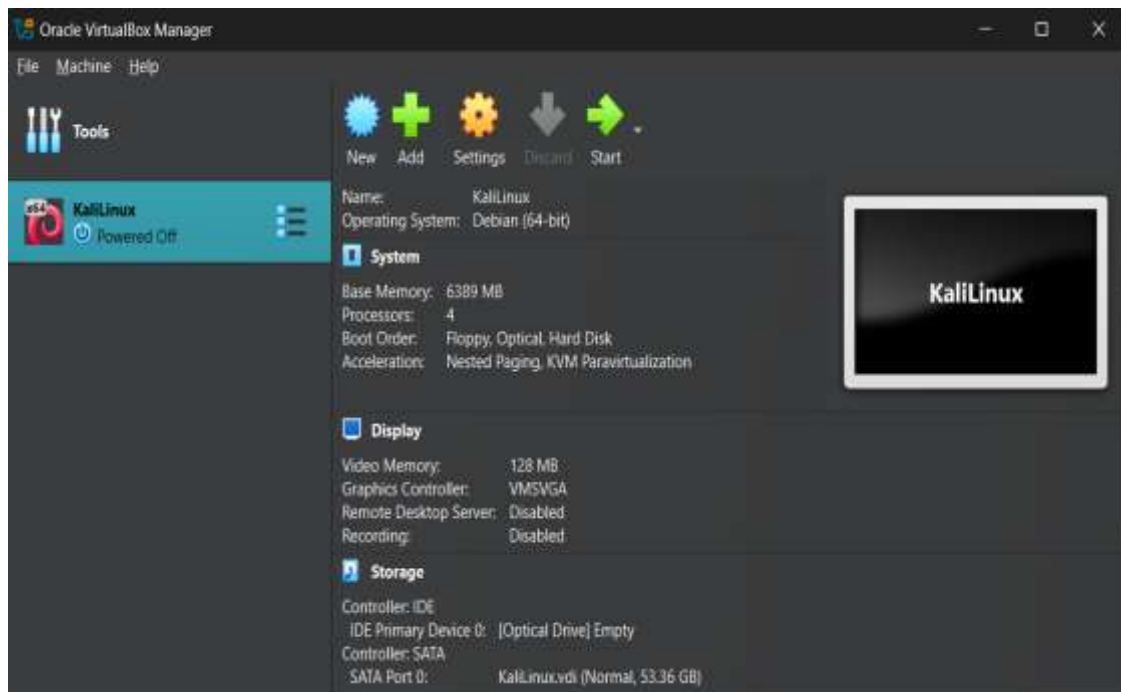
Installing Virtual Box:

Installing a virtual machine (VM) enables running multiple operating systems on one physical machine parallelly, useful for testing, development in isolated environments, and efficient resource use. VMs also support backups, disaster recovery, and enhanced security compared to traditional setups.



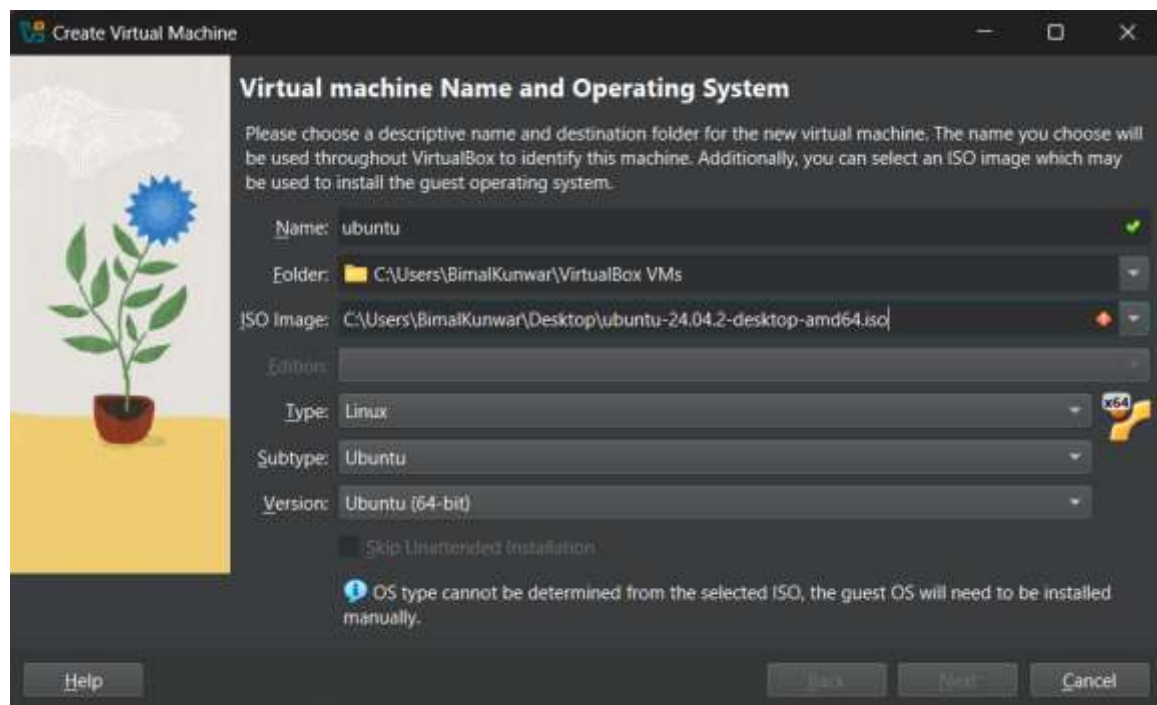
Here, click on **Next** and go to next phase.

Once Virtual Box is installed then we can create, add or import new VMs using toolbar buttons

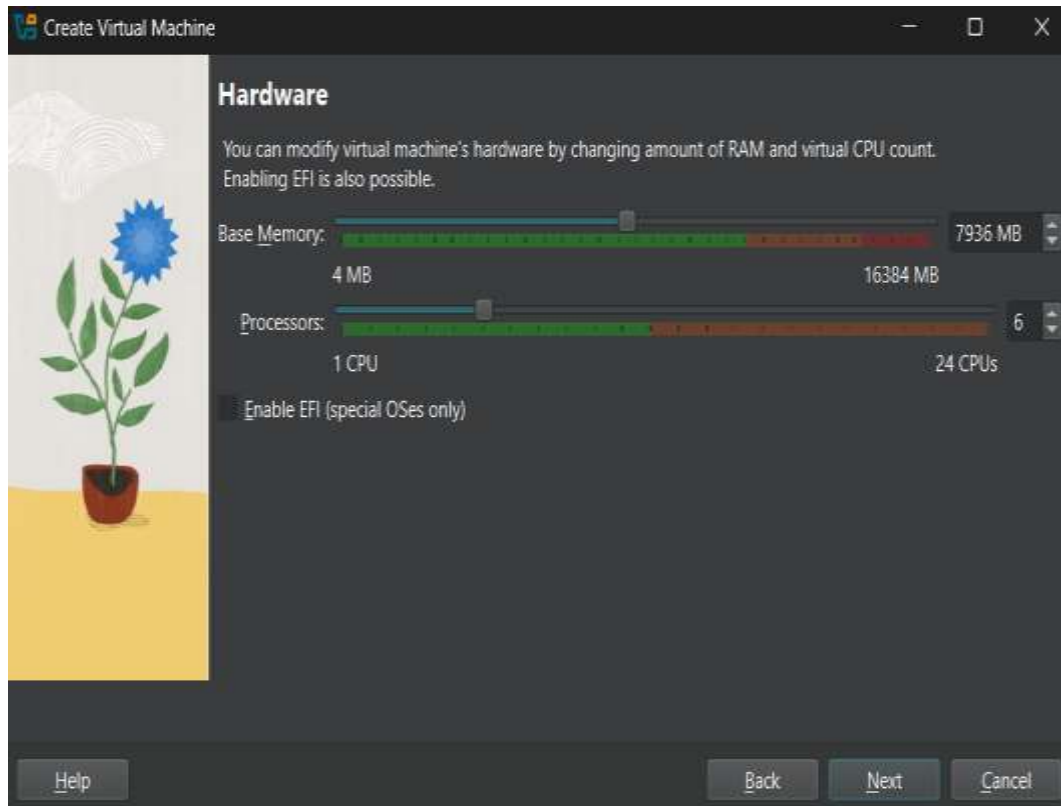


Installing Linux (Ubuntu) Desktop OS:

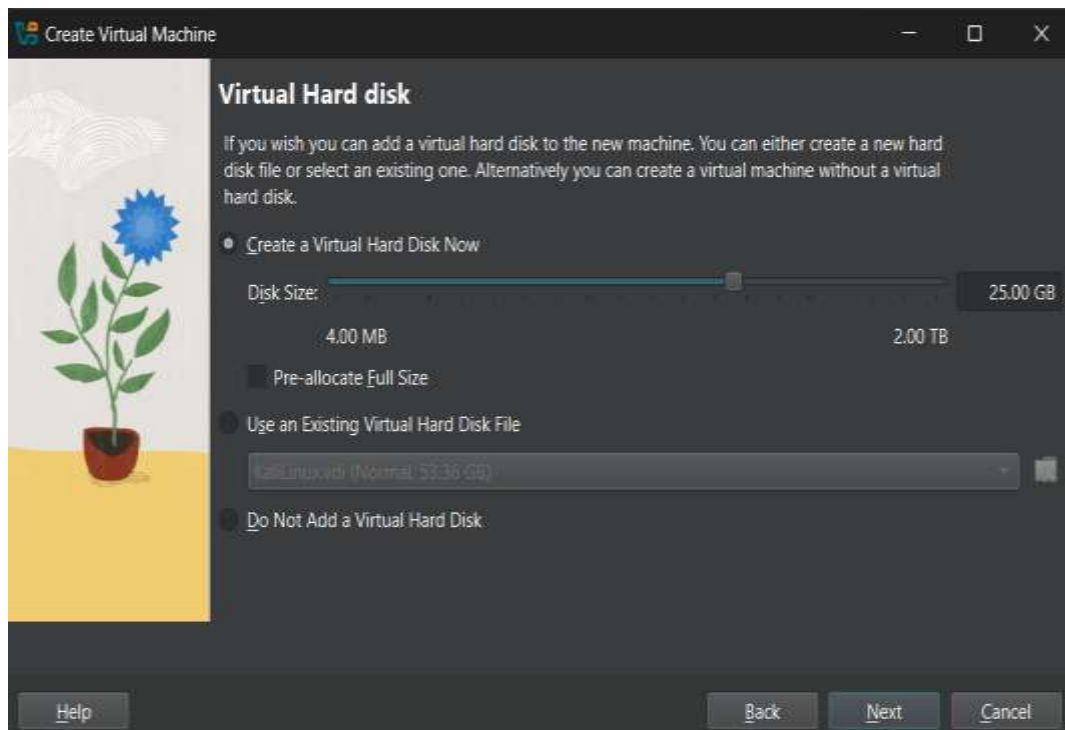
Step 1: Click New on Virtual Machine toolbar and add details about the new VM and required OS.



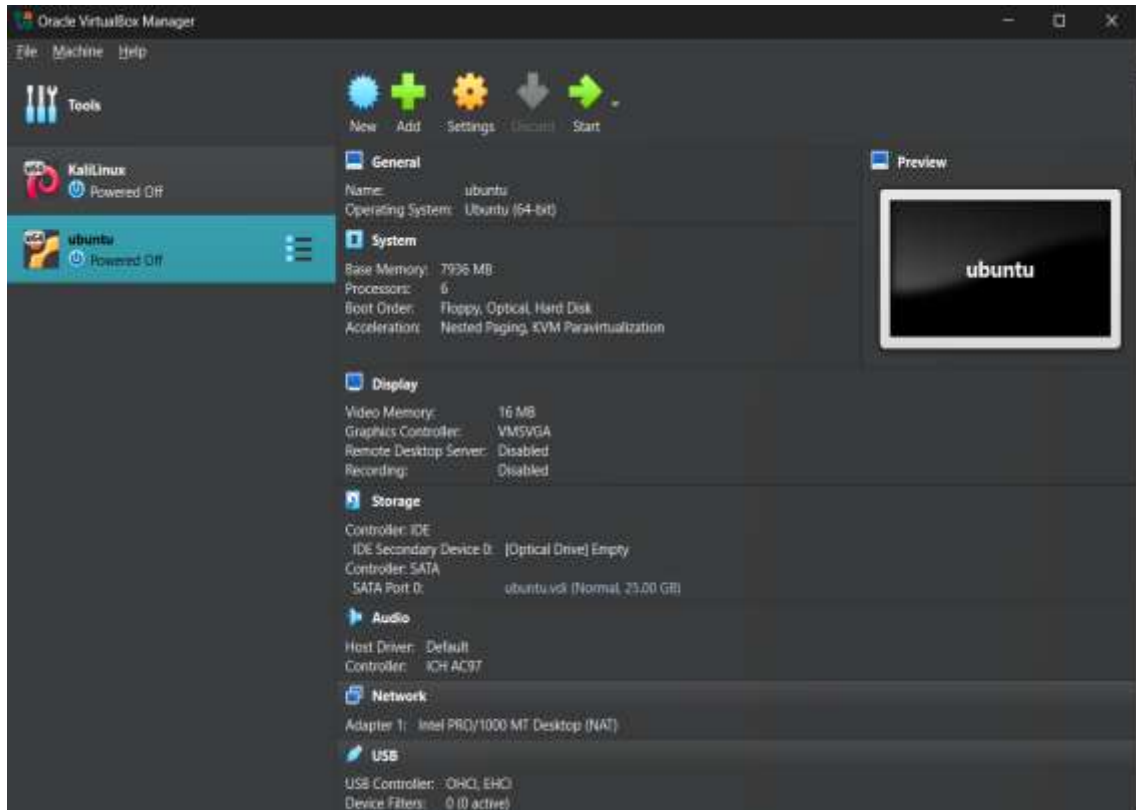
Step 2: Select the Base Memory and Processors to be allocated to the Virtual Machine as per your need.



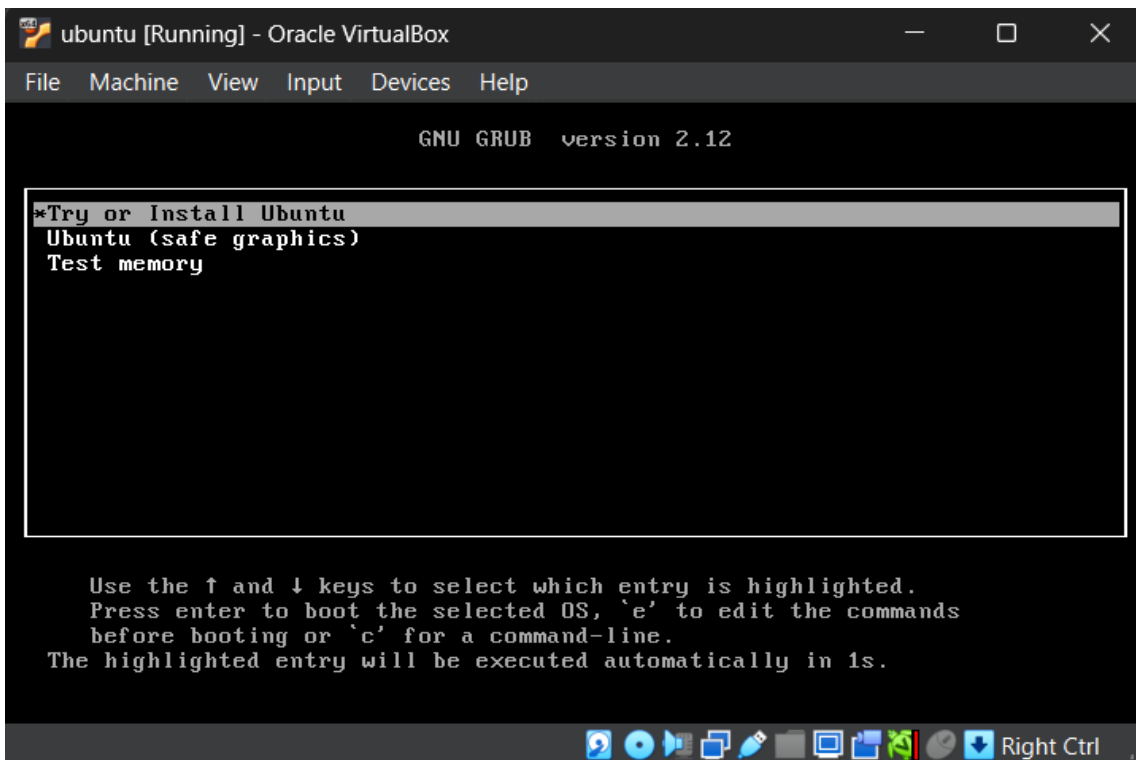
Step 3: Select the size of memory you want for Virtual Hard Disk.



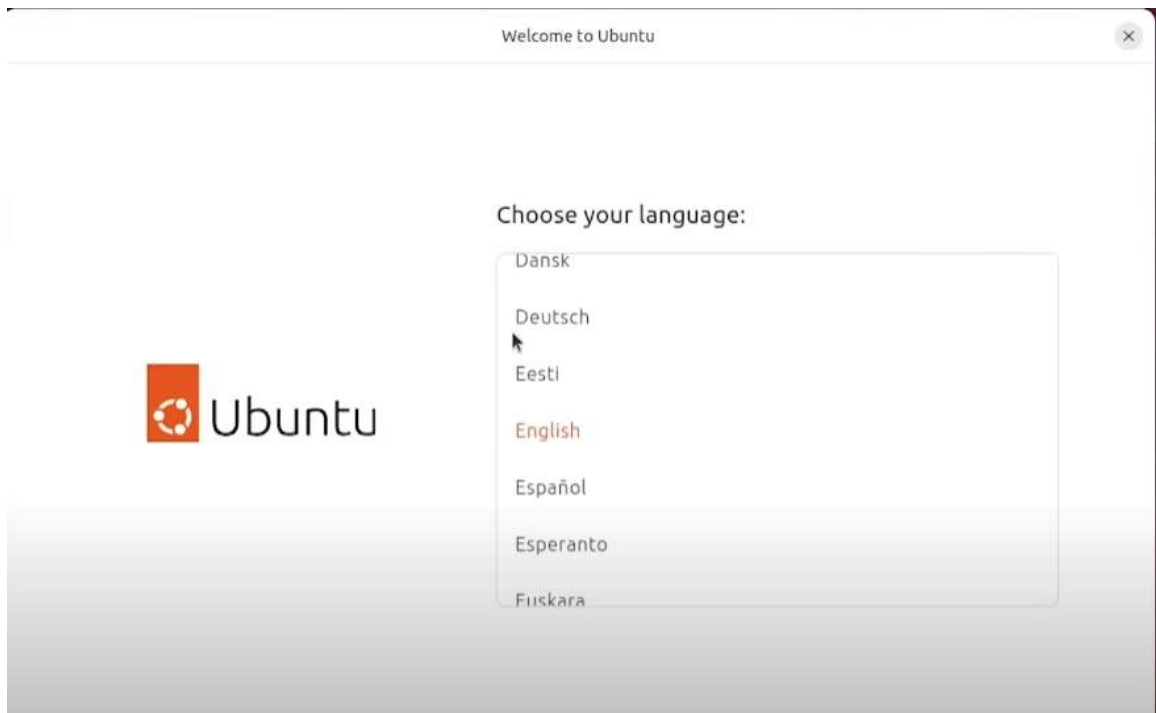
Step 4: This is the interface for the newly created Virtual Machine for Ubuntu Desktop OS. Select Server and click on Start to run the Virtual Machine



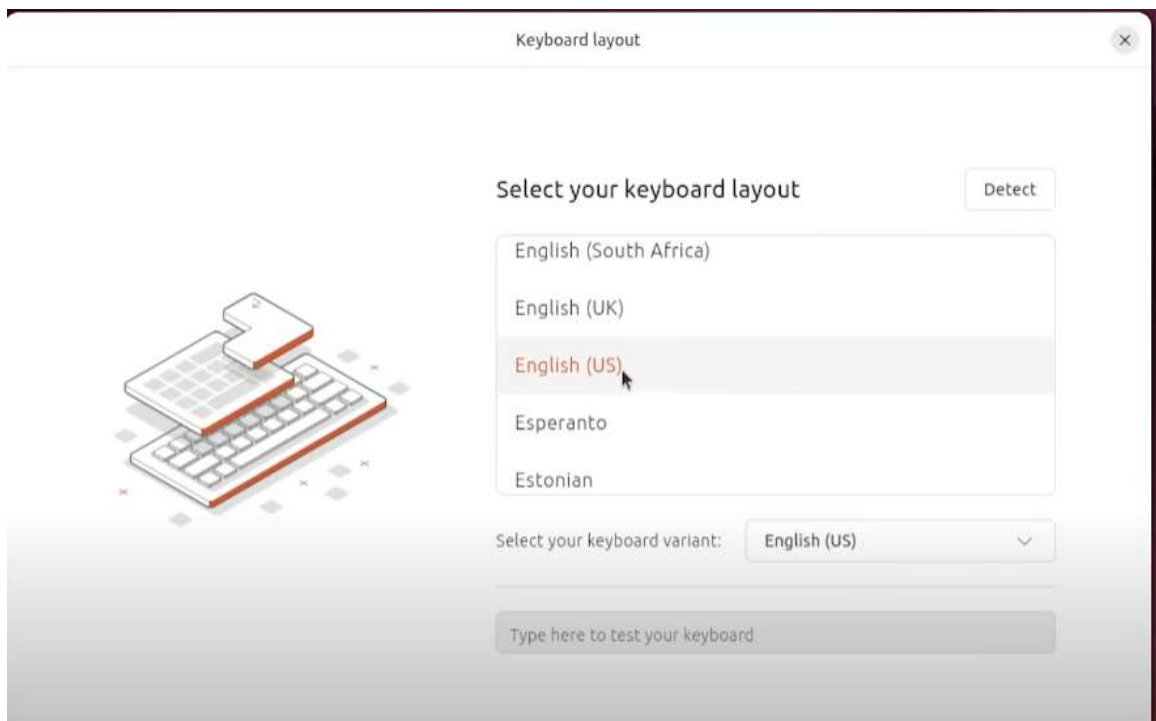
Step 5: Press Enter to install Ubuntu Server.



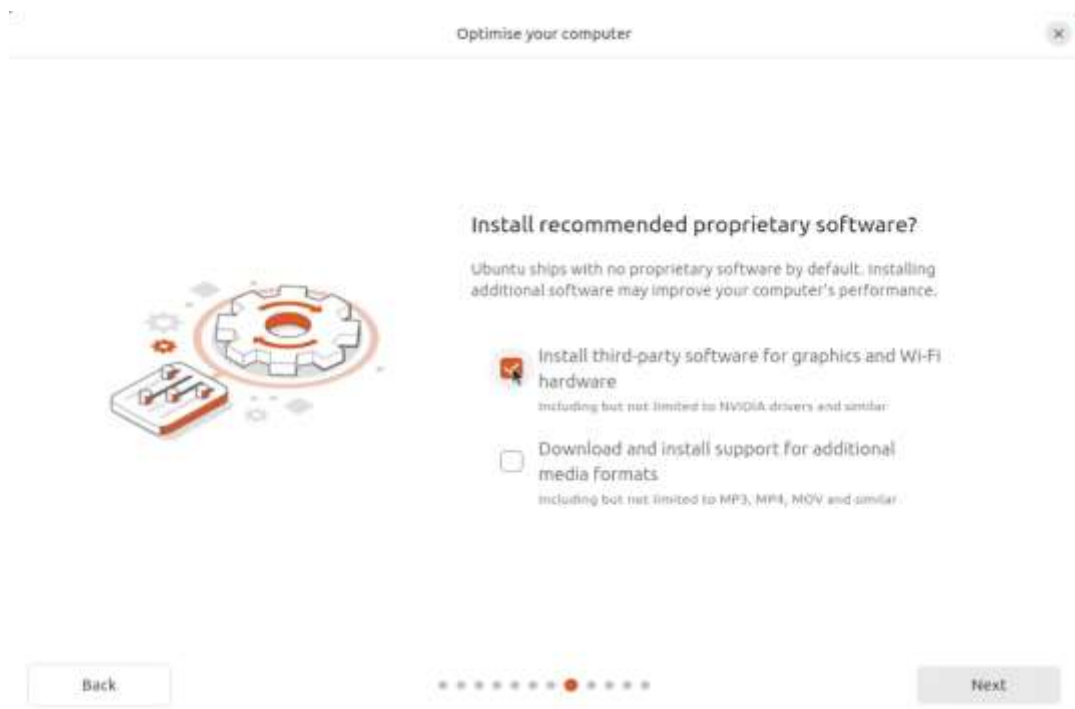
Step 6: Select the preferred language and click on Install Ubuntu.



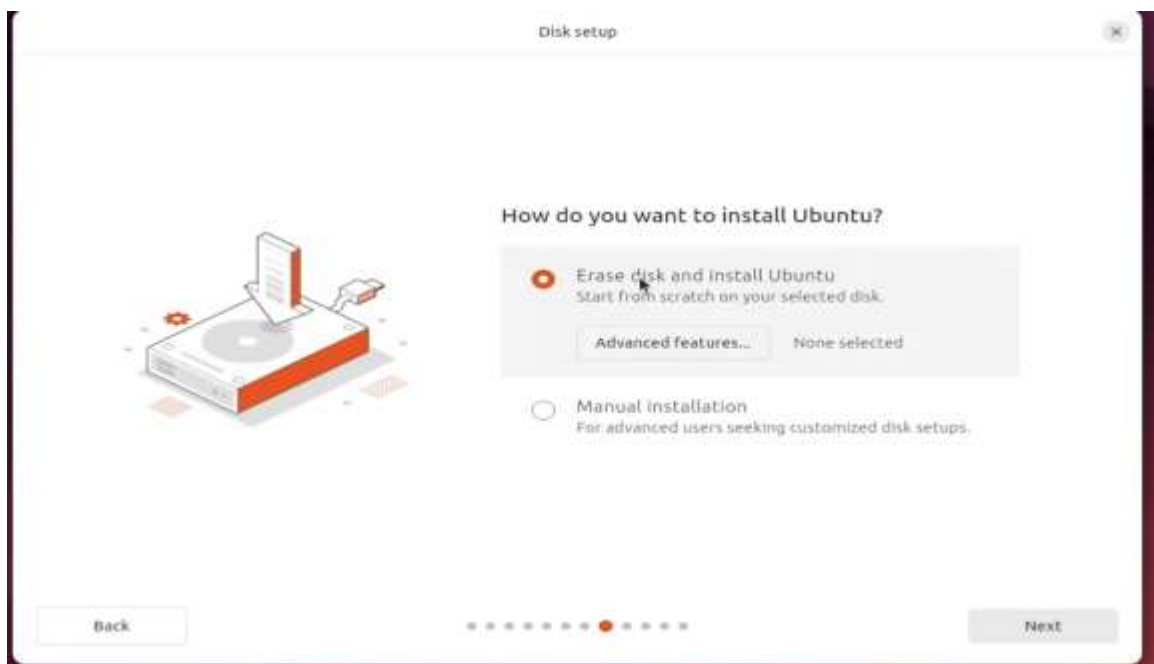
Step 7: Select the preferred keyboard layout and its variant, click on Continue once finished.



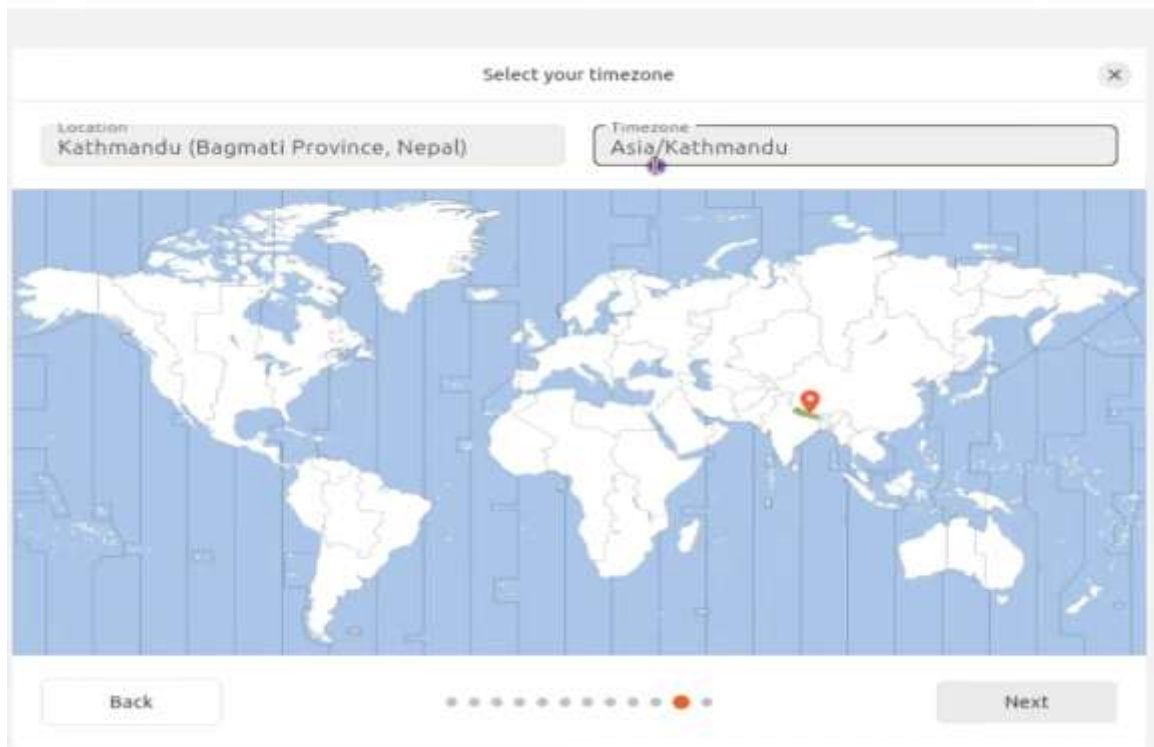
Step 8: It will prompt us to choose install recommended proprietary software to install alongside Ubuntu. We choose the options and click on Continue.



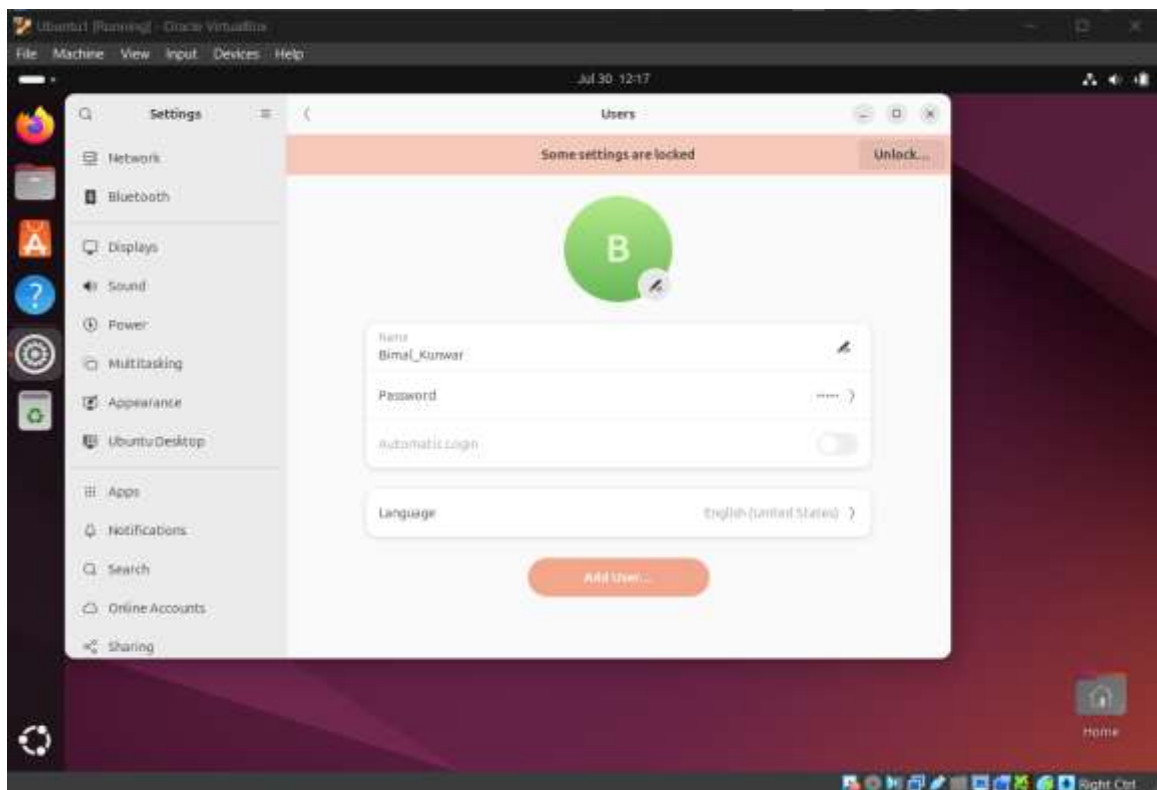
Step 9: We will be asked to choose the Installation type. Once selected we can press Enter. We will be informed about the partitions being formatted. If it's preferred, we press on Continue and proceed.



Step 10: Select the location and press Continue.

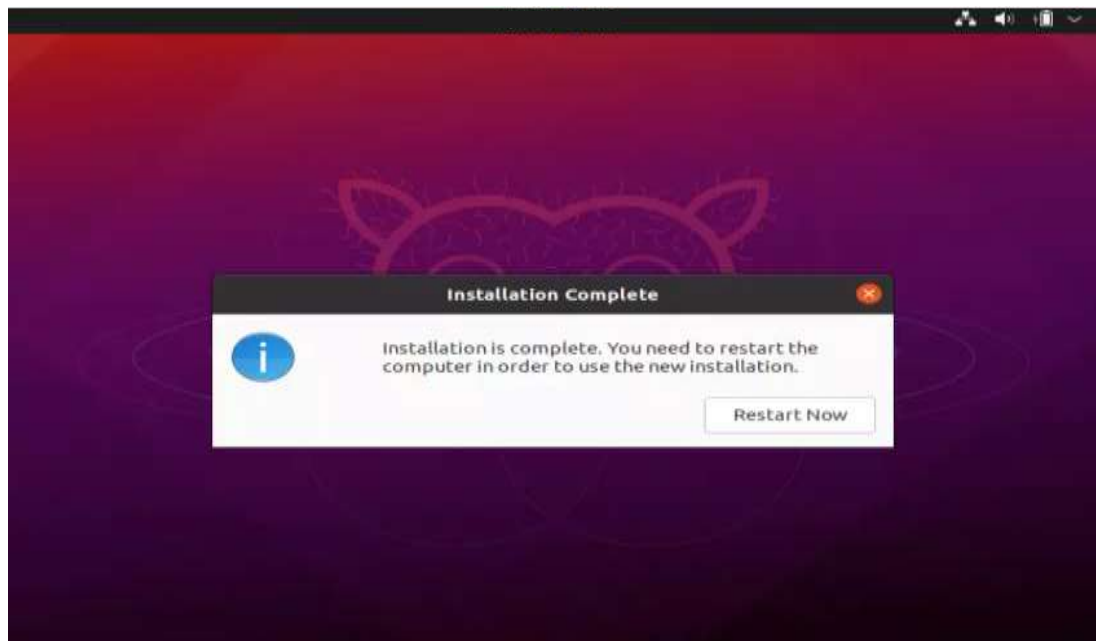


Step 11: Setup the profile for the user that includes username and password, once finished press Enter.

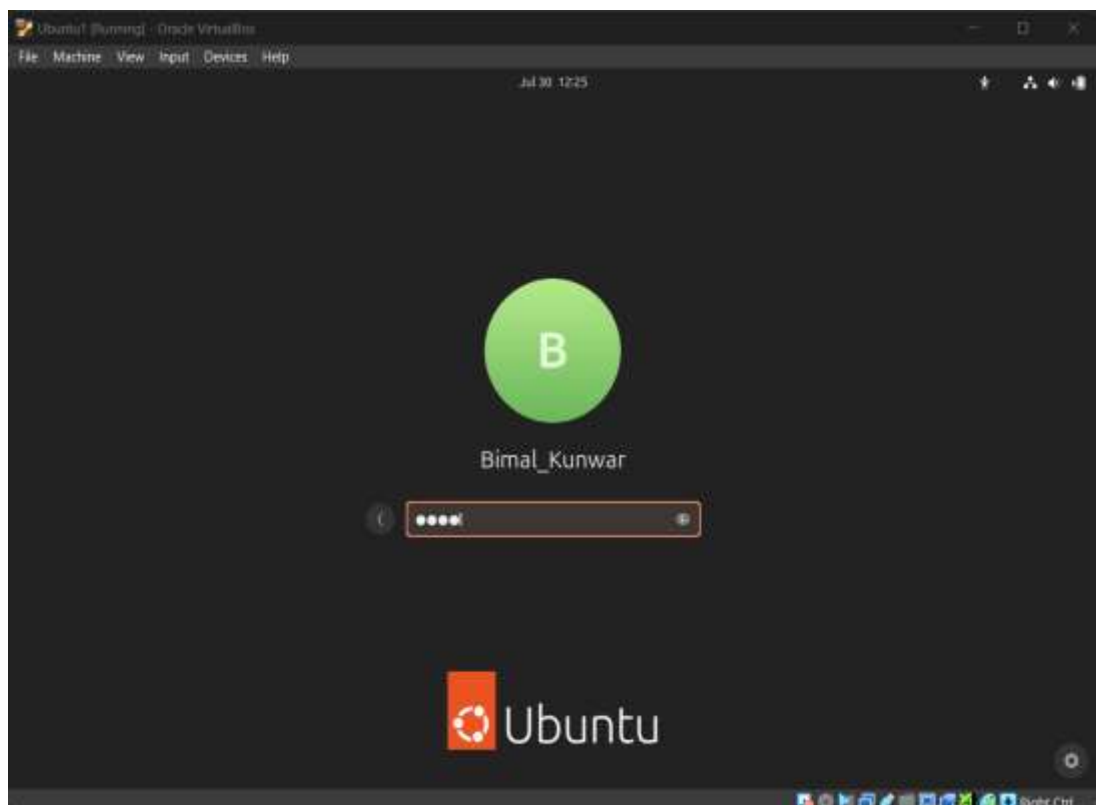


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Step 12: Upon the completion of the installation process, the window on the right will prompt, Restart Now.



Step 13: Now enter the Login details on the screen and press Enter to login into the system.



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Step 14: Now our system is ready and Ubuntu is installed completely.



Networking Commands:

1. PING

Definition: Ping (Packet Internet Groper) is a network utility used to test the reachability of a host and measure the round-trip time for messages sent from the originating host to a destination computer that are echoed back to the source.

Syntax: *ping [options] hostname or IP address*

Use: It is used to check the connectivity between my computer and the server to diagnose the network issue.

```
Bimal Kunwar
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ping bimalkunwar.com.np

Pinging bimalkunwar.com.np [172.67.176.112] with 32 bytes of data:
Reply from 172.67.176.112: bytes=32 time=7ms TTL=58
Reply from 172.67.176.112: bytes=32 time=6ms TTL=58
Reply from 172.67.176.112: bytes=32 time=6ms TTL=58
Reply from 172.67.176.112: bytes=32 time=4ms TTL=58

Ping statistics for 172.67.176.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 7ms, Average = 5ms
```

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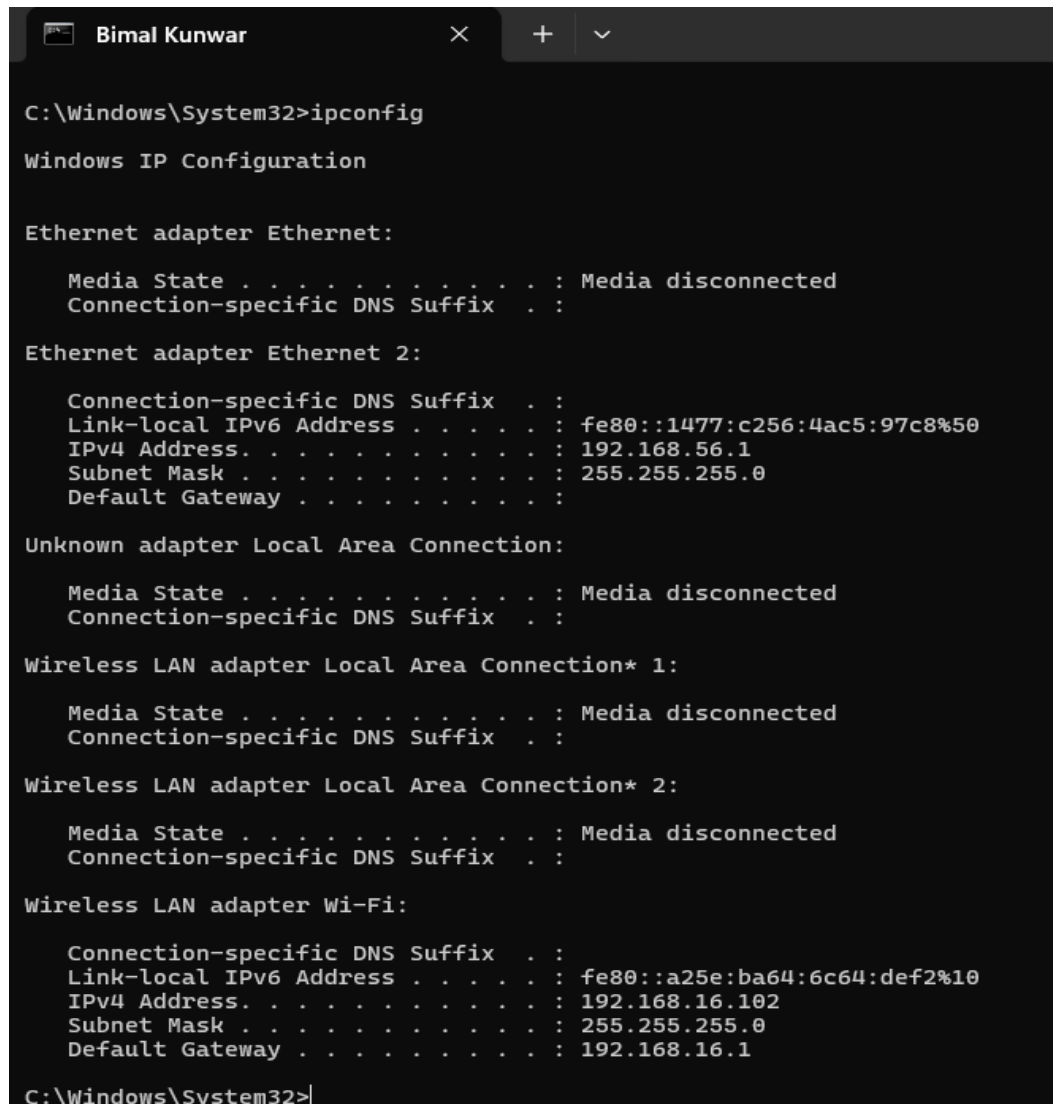
2. IPCONFIG

Definition: Ipconfig (Internet Protocol Configuration) is a command-line utility used to display and manage network configuration settings on Windows-based systems.

Syntax: Windows (ipconfig): *'ipconfig[/all]'*

Unix/Linux(ifconfig): *'ifconfig[interface]'*

Use: It is used to check the IP address assigned to my computer and verify the network settings before troubleshooting the connection issue.



```
C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::1477:c256:4ac5:97c8%50
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a25e:ba64:6c64:def2%10
    IPv4 Address. . . . . : 192.168.16.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.16.1

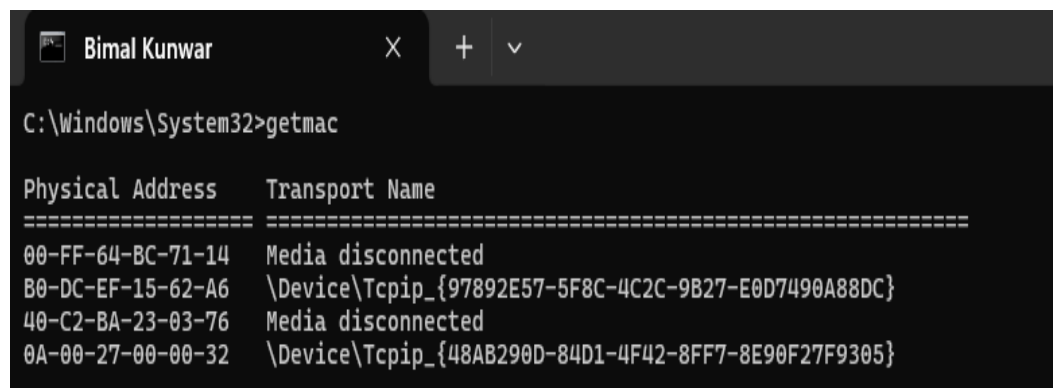
C:\Windows\System32>
```

3. GETMAC

Definition: GETMAC is a Windows command-line utility used to retrieve the Media Access Control (MAC) address for network adapters installed on a computer. The MAC address uniquely identifies each network interface card (NIC) and is used for network communication.

Syntax: *getmac*

Use: It is used to obtain the MAC addresses of all network adapters on my Windows machine, which helped in troubleshooting network connectivity issues and configuring specific network settings based on MAC address filtering.



```
Bimal Kunwar X + v
C:\Windows\System32>getmac

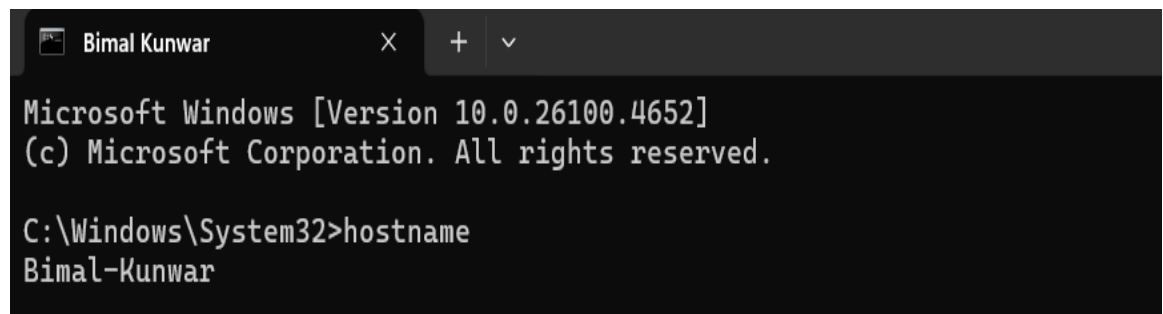
Physical Address    Transport Name
=====
00-FF-64-BC-71-14   Media disconnected
B0-DC-EF-15-62-A6   \Device\Tcpip_{97892E57-5F8C-4C2C-9B27-E0D7490A88DC}
40-C2-BA-23-03-76   Media disconnected
0A-00-27-00-00-32   \Device\Tcpip_{48AB290D-84D1-4F42-8FF7-8E90F27F9305}
```

4. HOSTNAME

Definition: A hostname is a label assigned to a device connected to a computer network that uniquely identifies it within that network.

Syntax: *hostname*

Use: To display the system name.



```
Bimal Kunwar X + v
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>hostname
Bimal-Kunwar
```

5. TRACERT

Definition: Tracert is a diagnostic utility that traces the path of a network packet from source to destination.

Syntax: *tracert <hostname>*

Use: It's a valuable tool for troubleshooting network issues like slow connections.

```
Bimal Kunwar
C:\Windows\System32>tracert bimalkunwar.com.np

Tracing route to bimalkunwar.com.np [172.67.176.112]
over a maximum of 30 hops:

  1    1 ms    1 ms    4 ms    blinkwifi.cn [192.168.16.1]
  2   199 ms   2 ms    4 ms    dsldevice.lan [192.168.1.254]
  3    6 ms    6 ms    3 ms    10.16.0.2
  4   41 ms    5 ms   12 ms    36.253.0.49
  5    8 ms    5 ms    8 ms   116.68.210.241
  6    4 ms    4 ms    5 ms   116.68.209.188
  7    7 ms    5 ms    6 ms   172.67.176.112

Trace complete.
```

6. NETSTAT

Definition: Netstat (NETwork STATistics) is a command-line tool used to display network connections, routing tables, interface statistics, and more.

Syntax: *netstat*

Use: To diagnose network issues by examining active connections, routing tables, and interface statistics on my Unix-based system.

```
Bimal Kunwar
C:\Windows\System32>netstat

Active Connections

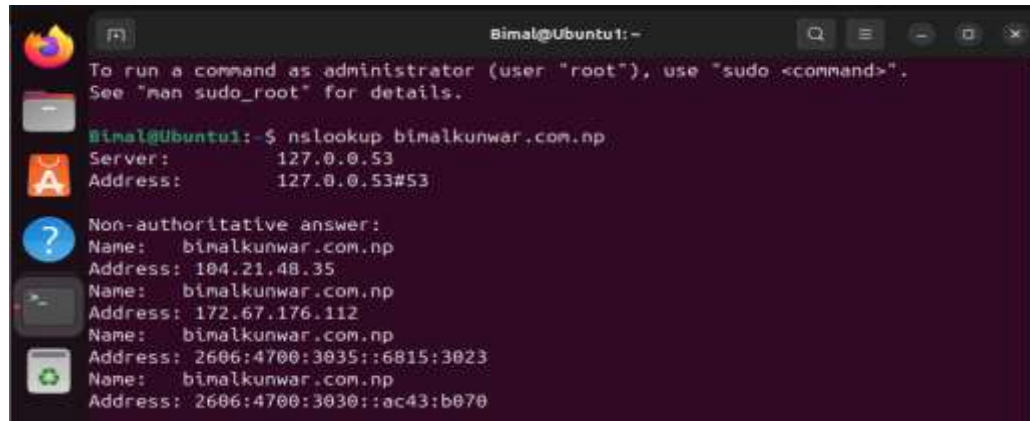
Proto Local Address           Foreign Address         State
TCP   127.0.0.1:49684          Bimal-Kunwar:49685     ESTABLISHED
TCP   127.0.0.1:49685          Bimal-Kunwar:49684     ESTABLISHED
TCP   127.0.0.1:49686          Bimal-Kunwar:49687     ESTABLISHED
TCP   127.0.0.1:49687          Bimal-Kunwar:49686     ESTABLISHED
TCP   127.0.0.1:49689          Bimal-Kunwar:49690     ESTABLISHED
TCP   127.0.0.1:49690          Bimal-Kunwar:49689     ESTABLISHED
TCP   127.0.0.1:49711          Bimal-Kunwar:49712     ESTABLISHED
TCP   127.0.0.1:49712          Bimal-Kunwar:49711     ESTABLISHED
TCP   127.0.0.1:49788          Bimal-Kunwar:49801     ESTABLISHED
TCP   127.0.0.1:49792          Bimal-Kunwar:49800     ESTABLISHED
TCP   127.0.0.1:49800          Bimal-Kunwar:49792     ESTABLISHED
TCP   127.0.0.1:49801          Bimal-Kunwar:49788     ESTABLISHED
TCP   127.0.0.1:49929          Bimal-Kunwar:49930     ESTABLISHED
TCP   127.0.0.1:49930          Bimal-Kunwar:49929     ESTABLISHED
TCP   127.0.0.1:49931          Bimal-Kunwar:49932     ESTABLISHED
TCP   127.0.0.1:49932          Bimal-Kunwar:49931     ESTABLISHED
TCP   127.0.0.1:49946          Bimal-Kunwar:49947     ESTABLISHED
TCP   127.0.0.1:49947          Bimal-Kunwar:49946     ESTABLISHED
TCP   127.0.0.1:49948          Bimal-Kunwar:49949     ESTABLISHED
TCP   127.0.0.1:49949          Bimal-Kunwar:49948     ESTABLISHED
TCP   192.168.16.102:50509      4.213.25.240:https     ESTABLISHED
TCP   192.168.16.102:50933      ec2-52-0-252-41:4244   ESTABLISHED
TCP   192.168.16.102:50937      155.133.225.21:27019   ESTABLISHED
```

7. NSLOOKUP

Definition: NSLOOKUP (Name Server Lookup) is a command-line tool used for querying Domain Name System (DNS) servers to obtain DNS records and information about domain names, IP addresses, and related network services.

Syntax: *nslookup <domain>*

Use: It is used for troubleshooting DNS issues, verifying DNS records, testing DNS configurations, and performing reverse DNS lookups.



```
Bimal@Ubuntu1: ~$ nslookup binalkunwar.com.np
Server:          127.0.0.53
Address:         127.0.0.53#53

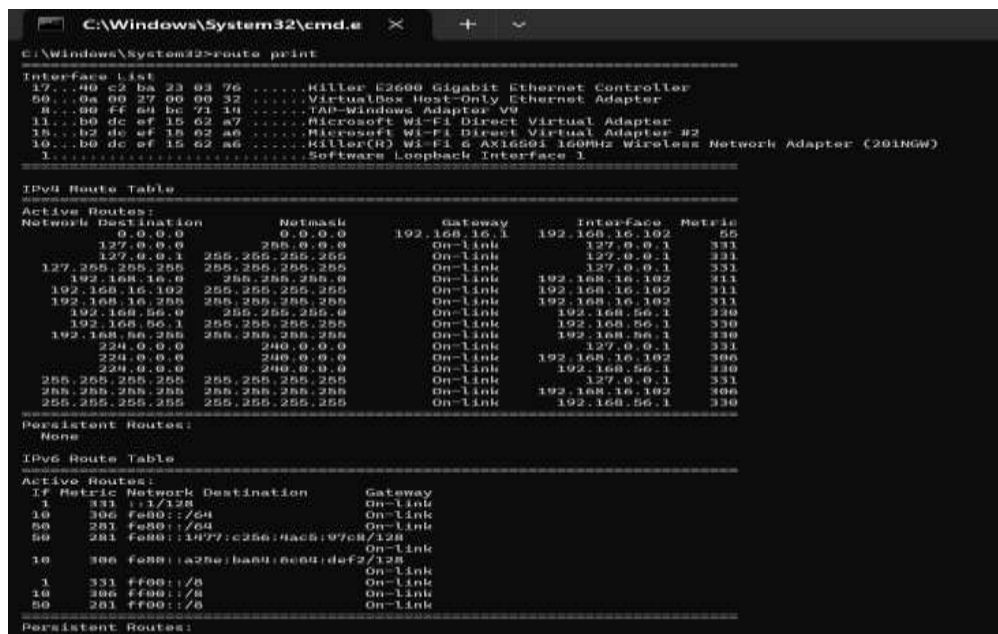
Non-authoritative answer:
Name:   binalkunwar.com.np
Address: 104.21.48.35
Name:   binalkunwar.com.np
Address: 172.67.176.112
Name:   binalkunwar.com.np
Address: 2606:4700:3035::6815:3023
Name:   binalkunwar.com.np
Address: 2606:4700:3030::ac43:b070
```

8. ROUTE PRINT

Definition: Route print refers to a command used to display the routing table of a device outlining the path that network traffic will take to reach various destinations.

Syntax: *route print*

Use: It shows the active network routes and how the system will direct traffic.



```
C:\Windows\System32\cmd.exe
C:\Windows\System32>route print

Interface List
17...40 c2 b5 23 03 76 .....Killer E2600 Gigabit Ethernet Controller
50...0a 00 27 00 00 32 .....VirtualBox Host-Only Ethernet Adapter
8...80 ff 64 bc 71 10 .....TAP-Windows Adapter V9
11...50 dc ef 15 62 a7 .....Microsoft Wi-Fi Direct Virtual Adapter
15...b2 dc ef 15 62 a6 .....Microsoft Wi-Fi Direct Virtual Adapter #2
10...b0 dc ef 15 62 a6 .....Killer(R) Wi-Fi 6 AX1650i 160MHz Wireless Network Adapter (201NGW)
1.....Software Loopback Interface 1

IPv4 Route Table
Active Routes:
Network Destination        Netmask          Gateway           Interface         Metric
0.0.0.0                    0.0.0.0          192.168.16.1      192.168.16.102    55
127.0.0.0                  255.0.0.0        On-link          127.0.0.1         331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1         331
127.255.255.255            255.255.255.255  On-link          127.0.0.1         331
192.168.16.0               255.255.255.0    On-link          192.168.16.102    311
192.168.16.102             255.255.255.255  On-link          192.168.16.102    311
192.168.16.255             255.255.255.255  On-link          192.168.16.102    311
192.168.56.0               255.255.255.0    On-link          192.168.56.1      330
192.168.56.1               255.255.255.255  On-link          192.168.56.1      330
224.0.0.0                  240.0.0.0        On-link          127.0.0.1         331
224.0.0.0                  240.0.0.0        On-link          192.168.16.102    306
224.0.0.0                  240.0.0.0        On-link          192.168.56.1      330
255.255.255.255            255.255.255.255  On-link          127.0.0.1         331
255.255.255.255            255.255.255.255  On-link          192.168.16.102    306
255.255.255.255            255.255.255.255  On-link          192.168.56.1      330

Persistent Routes:
None

IPv6 Route Table
Active Routes:
If Metric Network Destination      Gateway
1 331 ::1/128 ::1 On-link
10 306 fe80::/64 fe80:: On-link
50 281 fe80::/64 fe80:: On-link
50 281 fe80::1077:c266:4ac6:97c8/128 fe80:: On-link
10 306 fe80::a20e:b800:ec00:def2/128 fe80:: On-link
1 331 ff00::/8 ff00:: On-link
10 306 ff00::/8 ff00:: On-link
50 281 ff00::/8 ff00:: On-link

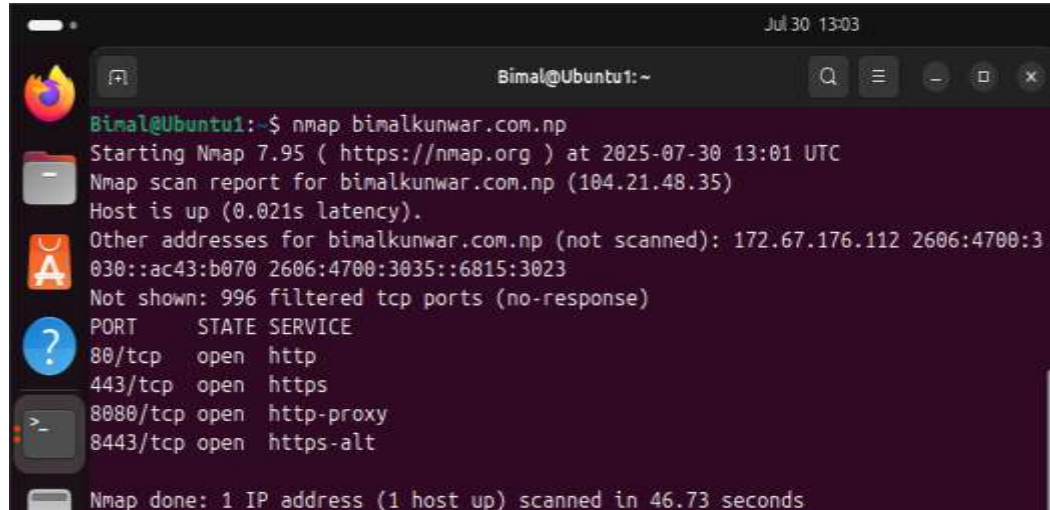
Persistent Routes:
None
```

9. NMAP

Definition: Nmap (Network Mapper) is a powerful, free, and open-source tool used for network discovery and security auditing.

Syntax: `nmap <target>`

Use: It helps admins and security professionals identify potential weaknesses.

A terminal window titled 'Bimal@Ubuntu1: ~' showing the output of an Nmap scan. The command executed is 'nmap bimalkunwar.com.np'. The output indicates the host is up with a latency of 0.021s. It lists other unscanned addresses and shows a table of open ports: 80/tcp (http), 443/tcp (https), 8080/tcp (http-proxy), and 8443/tcp (https-alt). The scan took 46.73 seconds.

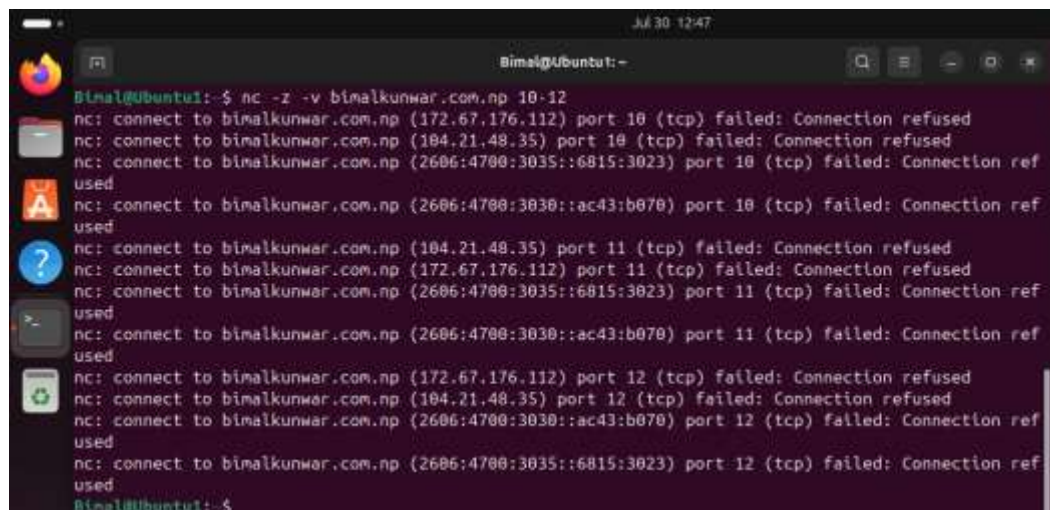
```
Bimal@Ubuntu1:~$ nmap bimalkunwar.com.np
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-30 13:01 UTC
Nmap scan report for bimalkunwar.com.np (104.21.48.35)
Host is up (0.021s latency).
Other addresses for bimalkunwar.com.np (not scanned): 172.67.176.112 2606:4700:3
030::ac43:b070 2606:4700:3035::6815:3023
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
Nmap done: 1 IP address (1 host up) scanned in 46.73 seconds
```

10. NETCAT

Definition: Netcat is a versatile command-line tool used in computer networking to read and write data across network connections, supporting both TCP and UDP protocols

Syntax: `nc -z -v <target> <port>`

Use: It enables actions like establishing connections, transferring data, and even basic port scanning.

A terminal window titled 'Bimal@Ubuntu1: ~' showing the output of a Netcat port scan. The command executed is 'nc -z -v bimalkunwar.com.np 10-12'. The output shows multiple connection attempts to various IP addresses and ports (10, 11, 12) for different hosts, all resulting in 'Connection refused'.

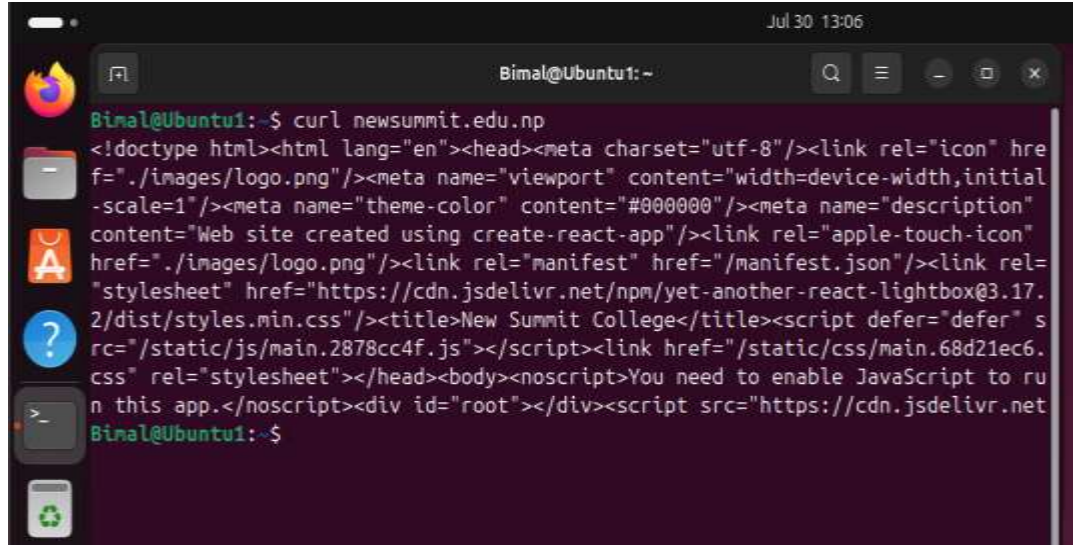
```
Bimal@Ubuntu1:~$ nc -z -v bimalkunwar.com.np 10-12
nc: connect to bimalkunwar.com.np (172.67.176.112) port 10 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (104.21.48.35) port 10 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (2606:4700:3035::6815:3023) port 10 (tcp) failed: Connection ref
used
nc: connect to bimalkunwar.com.np (2606:4700:3030::ac43:b070) port 10 (tcp) failed: Connection ref
used
nc: connect to bimalkunwar.com.np (104.21.48.35) port 11 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (172.67.176.112) port 11 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (2606:4700:3035::6815:3023) port 11 (tcp) failed: Connection ref
used
nc: connect to bimalkunwar.com.np (2606:4700:3030::ac43:b070) port 11 (tcp) failed: Connection ref
used
nc: connect to bimalkunwar.com.np (172.67.176.112) port 12 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (104.21.48.35) port 12 (tcp) failed: Connection refused
nc: connect to bimalkunwar.com.np (2606:4700:3030::ac43:b070) port 12 (tcp) failed: Connection ref
used
nc: connect to bimalkunwar.com.np (2606:4700:3035::6815:3023) port 12 (tcp) failed: Connection ref
used
Bimal@Ubuntu1:~$
```

11. CURL

Definition: Curl (Client URL) is a command line tool that developers use to transfer data to and from a server

Syntax: `curl <URL>`

Use: It is used to transfer data to and from Internet servers.

A screenshot of a terminal window on an Ubuntu system. The window title is "Bimal@Ubuntu1: ~". The terminal shows a command prompt where the user has entered `curl newsummit.edu.np`. The output is an HTML document. The first line is `<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href=`. The second line is `="/images/logo.png"/><meta name="viewport" content="width=device-width,initial`. The third line is `-scale=1"/><meta name="theme-color" content="#000000"/><meta name="description"`. The fourth line is `content="Web site created using create-react-app"/><link rel="apple-touch-icon"`. The fifth line is `href="/images/logo.png"/><link rel="manifest" href="/manifest.json"/><link rel=`. The sixth line is `"stylesheet" href="https://cdn.jsdelivr.net/npm/yet-another-react-lightbox@3.17.`. The seventh line is `2/dist/styles.min.css"/><title>New Summit College</title><script defer="defer" s`. The eighth line is `rc="/static/js/main.2878cc4f.js"></script><link href="/static/css/main.68d21ec6.`. The ninth line is `css" rel="stylesheet"></head><body><noscript>You need to enable JavaScript to ru`. The tenth line is `n this app.</noscript><div id="root"></div><script src="https://cdn.jsdelivr.net`. The terminal prompt `Bimal@Ubuntu1:~$` is visible at the bottom.

12. SYSTEMINFO

Definition: Displays detailed configuration information about a computer and its operating system, including operating system configurations, etc.

Syntax: `systeminfo`

Use: It helps us know hardware properties, product ID, security information, and so on.

```
Bimal Kunwar
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>systeminfo

Host Name:                               BIMAL-KUNWAR
OS Name:                                 Microsoft Windows 11 Home Single Language
OS Version:                             10.0.26100 N/A Build 26100
OS Manufacturer:                         Microsoft Corporation
OS Configuration:                       Standalone Workstation
OS Build Type:                            Multiprocessor Free
Registered Owner:                         Acer
Registered Organization:                  N/A
Product ID:                               00356-24602-22654-AAOEM
Original Install Date:                    12/14/2024, 8:56:52 PM
System Boot Time:                         7/30/2025, 2:43:43 PM
System Manufacturer:                      Acer
System Model:                             Nitro AN515-58
System Type:                              x64-based PC
Processor(s):                             1 Processor(s) Installed.
[01]: Intel(R) Family 6 Model 164 Stepping 8 GenuineIntel ~3100 Mhz
BIOS Version:                             Insyde Corp. V2.17, 2/20/2025
Windows Directory:                       C:\WINDOWS
System Directory:                         C:\WINDOWS\system32
Boot Device:                              \Device\HarddiskVolume1
System Locale:                             en-us;English (United States)
Input Locale:                             en-us;English (United States)
Time Zone:                                (UTC+05:45) Kathmandu
Total Physical Memory:                     16,088 MB
Available Physical Memory:                 3,672 MB
Virtual Memory: Max Size:                 25,997 MB
Virtual Memory: Available:                1,290 MB
Virtual Memory: In Use:                   33,807 MB
Page File Location(s):                    C:\pagefile.sys
Domain:                                   WORKGROUP
Logon Server:                             \\BIMAL-KUNWAR
Hotfix(s):                                 2 Hotfix(s) Installed.
[01]: KB5056579
[02]: KB5062983
[03]: KB5063668
4 NIC(s) Installed.
[01]: TAP-Windows Adapter V9
Connection Name: Local Area Connection
Status: Media disconnected
[02]: Killer(R) Wi-Fi 6 AX1600i 160MHz Wireless Network Adapter (2019GW)
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 192.168.16.1
IP address(es)
[01]: 192.168.16.102
[02]: fe80::a25e:ba84:6c64:def2
[03]: Killer E2600 Gigabit Ethernet Controller
Connection Name: Ethernet
Status: Media disconnected
[04]: VirtualBox Host-Only Ethernet Adapter
Connection Name: Ethernet 2
DHCP Enabled: No
IP address(es)
[01]: 192.168.56.1
[02]: fe80::1477:c256:4ac5:97c8
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMM Security Mitigations 1.0
Mode Based Execution Control
APIC Virtualization
Services Configured:
Services Running:
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

Conclusion:

In this practical, the lab work involving the installation of VirtualBox or VMware and Ubuntu OS provided valuable insights into operating system fundamentals and virtualization concepts and technologies. Through practical exercises with basic networking commands such as PING, IPCONFIG, GETMAC, HOSTNAME, NSLOOKUP, TRACERT, NETSTAT, SYSTEMINFO, ROUTE PRINT, NMAP, NETCAT, and CURL, participants gained proficiency in network diagnostics, configuration, and system management. These activities contributed to a deeper understanding of essential tools for troubleshooting and optimizing network performances.